two questions, four boxes

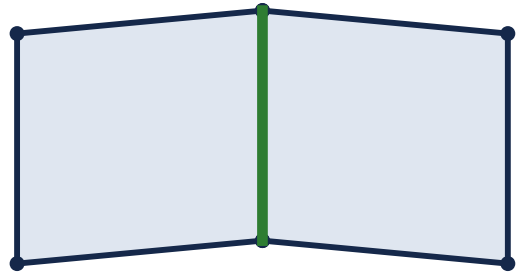
across: is the gap legal for the equation you are solving? down: is there a gap at all?

NO GAP

1. CONTINUOUS + CONFORMING

no gap, and none is wanted

one shared edge



Q4 in plane stress, heat, Poisson

the ordinary case --- nothing to think about

2. CONTINUOUS + NON-CONFORMING

no gap, but the equation wants more than that

deflection joins --- slope does not

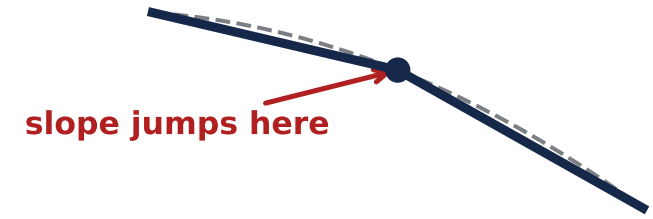


plate / beam elements: ACM (MZC), DKT

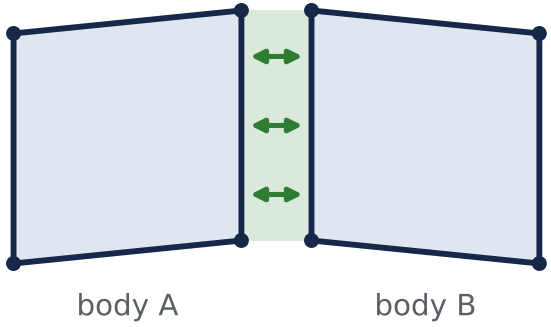
a plate equation needs the slope to match too

GAP

3. DISCONTINUOUS + CONFORMING

gap, but it sits on a boundary --- not inside a body

two domains, not one



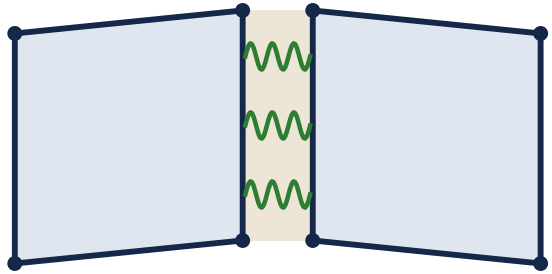
two bodies in contact; a cohesive crack

each piece is legal on its own domain

4. DISCONTINUOUS + NON-CONFORMING

gap inside one body --- so who pays for it?

one body, torn on purpose



penalty term present ⇒ DG. Works.

nothing present ⇒ Wilson Q6 on distorted quads, Morley. Fails the patch test.